

JÖNKÖPING UNIVERSITY
School of Engineering

Autonomous robot for warehouse cleaning

MAIN AREA: Product development & design

AUTHORS: Carl Holmqvist, Noa Rönning

SUPERVISOR: Peter Andersson

EXAMINER: Marisol Rico Cortez

JÖNKÖPING 2024-12-16 - 2025-05-30

Detta examensarbete är utfört vid Tekniska Högskolan i Jönköping inom Maskinteknik: Produktutveckling och design. Författarna svarar själva för framförda åsikter, slutsatser och resultat.

Examinator: Marisol Rico Cortez
Handledare: Peter Andersson
Omfattning: 15 hp (grundnivå)
Datum: 2025-05-30

Postadress:	Besöksadress:	Telefon:
Box 1026	Gjuterigatan 5	036-10 10 00 (vx)
551 11 Jönköping		

Abstract

This project presents a conceptual design for an autonomous cleaning robot intended for warehouse environments, designed as a potential future product for the company Husqvarna. It was completed in collaboration with Semcon/Knightec Group, aiming to explore how Husqvarna could enter a new market with a product that communicates the company's core values of sustainability, quality and innovation.

The project followed the Double Diamond design process, including interviews, market analysis, ideation, 3D-modelling, renderings and prototyping. The robot, called Pura, features a modular system that allows it to switch between different cleaning modules to handle both large open areas and hard-to-reach spaces. It operates fully autonomously, including automatic charging and trash disposal, and is designed to reduce the physical strain for warehouse cleaners, caused by manual cleaning.

The final concept reflects Husqvarna's established visual identity while introducing innovative functions suited for professional use in warehouse environments. The design is communicated primarily through detailed digital renderings that highlight form, function and usability, aligned with the brand's design language. The project demonstrates Husqvarna's potential to expand into the warehouse cleaning industry with an innovative idea that focuses on user needs.

Keywords: industrial design, product development, Husqvarna, autonomous robot, warehouse cleaning

Sammanfattning

Detta projekt presenterar ett designkoncept för en autonom städrobot avsedd för lagermiljöer, utvecklad som en möjlig framtida produkt för företaget Husqvarna. Arbetet genomfördes i samarbete med Semcon/Knitec Group med målet att undersöka hur Husqvarna kan etablera sig på en ny marknad med en produkt som förmedlar företagets formspråk, med fokus på hållbarhet, kvalitet och innovation.

Projekt följde Double Diamond designprocessen och omfattade intervjuer, marknadsanalys, idégenerering, 3D-modellering, renderingar och prototypframtagning. Den modulära roboten, med namnet Pura, kan växla mellan olika städmoduler för att hantera både stora öppna ytor och svåråtkomliga utrymmen. Den är fullt autonom och hanterar både laddning och tömning av skräp automatiskt, Designen syftar till att minska den fysiska belastningen för lagerpersonal som annars uppstår vid manuell rengöring.

Det slutgiltiga konceptet speglar Husqvarnas etablerade formspråk och introducerar samtidigt innovativa funktioner anpassade för professionell användning i industriella miljöer. Designen kommuniceras främst genom detaljerade digitala renderingar som lyfter fram form, funktion och användarvänlighet i linje med varumärkets design principer. Projektet visar på Husqvarnas potential att ta sig in på marknaden för lagerrengöring med en innovativ lösning som utgår från användarnas behov.

Table of content

1. Introduction 8
1.1 THE COMPANY 8
1.2 TARGET COMPANY 8
1.3 BACKGROUND 8
1.4 PURPOSE AND RESEARCH QUESTIONS 9
1.5 LIMITATIONS 9
1.6 DISPOSITION 9
2. Theoretical Framework 10
2.1 WHAT IS PRODUCT DEVELOPMENT 10
2.2 WHAT IS DESIGN 10
2.2.1 Industrial design 10
2.3 THE DESIGN PROCESS 11
2.3.1 Design Methods 11
2.3.2 The double diamond process 11
2.4 FORM LANGUAGE 11
2.5 VACUUM CLEANER 12
2.6 ROBOT FLOOR CLEANER 12
2.6.1 History 12
2.6.2 Navigation Technology 12
2.7 ROBOT FLOOR CLEANER CONSIDERATIONS 13
2.7.1 Floor cleaning process in warehouse 13
2.7.2 Robot Size 14
3. Method 15
3.1 IDENTIFYING CUSTOMER NEEDS 15
3.1.1 Market research 15
3.1.2 Target audience analysis 16
3.2 PRODUCT SPECIFICATIONS 16
3.2.1 Product MoSCoW 16
3.3 CONCEPT GENERATION 17
3.3.1 Brainstorming 17
3.4 CONCEPT SELECTION 17
3.5 CONCEPT TESTING 18

3.5.1 Computer-aided design 18
3.5.2 Render 18
3.5.3 Prototyping 18
4. Approach & Implementation 19
4.1 PRODUCT BOARD 19
4.2 TREND ANALYSIS 19
4.3 INTERVIEW 20
4.4 PERSONAS 21
4.5 MOOD BOARD 22
4.6 MOSCOW 23
4.7 BRAINSTORMING 24
4.8 CONCEPT IDEATION 25
4.9 MID-PRESENTATION DESIGN CONCEPTS 26
4.9.1 Concept 1 27
4.9.2 Concept 2 28
4.9.3 Concept 3 29
4.9.4 Concept 4 30
4.9.5 Concept 5 31
4.10 DESIGN UPDATE 32
4.11 DIMENSION TESTING 33
4.12 FINAL DESIGN SKETCH 34
4.13 SURFACE MODELLING 35
4.14 PROTOTYPE 37
5. Result 38
5.1 DESIGN 38
5.2 PRODUCT NAME 39
5.3 PRODUCT FEATURES 40
5.3.1 Cleaning function 40
5.3.2 Modularity 41
5.3.3 Charging 42
5.3.4 Trash disposal 43
5.3.5 Control panel 44
5.3.6 Navigation 45
5.3.7 Position light 46

5.4 MULTICOLOUR OPTIONS 47
5.5 PROTOTYPE 48
6. Conclusion and discussion 49
6.1 RESEARCH QUESTIONS 49
6.1.1 Husqvarna's product identity, Research questions 1 49
6.1.2 Husqvarna robotic vacuum cleaner, Research question 2 49
6.2 USED METHODS 50
6.3 FURTHER DEVELOPMENT 51
6.4 CONCLUSION 52
7. References 53

1. Introduction

This report reviews the working methods of the design process for a robotic vacuum cleaner, designed using Husqvarna's values and design language. The robotic vacuum cleaner is adapted to function in a warehouse environment with necessary functions. The project is a collaboration between Jönköping University and Semcon/Knightec Group to create a product targeted towards Husqvarna.

1.1 The company

Semcon/Knightec Group is the company leading this project. The company is an international technology partner specializing in engineering services and product information solutions. Semcon was founded in Västerås in 1980 but relocated its headquarters to Gothenburg five years later. In 2024, Semcon merged with Knightec Group, becoming one of Northen Europe's leading product and digital service consultant firm, with approximately 2400 employees located in more than 20 locations [1].

1.2 Target company

Together with Semcon/Knightec Group this project is targeted towards Husqvarna. Husqvarna was founded in 1689 and started as a manufacturer of muskets for the Swedish royal army. Over a long time, Husqvarna has diversified its product range from weapons to sewing machines, bicycles, kitchen tools and chainsaws. Today, Husqvarna offers a wide range of equipment for forest, construction, park and garden care. Products are offered for both professional users and everyday consumers, with a strong focus on developing sustainable and innovative solutions [2].

1.3 Background

Cleaning solutions are no exception to how automation and smart technology are transforming industries. Keeping warehouse spaces clean is critical to the efficiency of operations, workplace safety, and long-lasting equipment. Because traditional cleaning techniques frequently involve intensive manual labour and time commitment, autonomous alternatives such as robotic vacuum cleaners are becoming increasingly a more efficient and attractive option.

The goal of this project is to use Husqvarna's design language and values to create a robotic vacuum cleaner for use in warehouses. Husqvarna maintains high standards due to its reputation for producing innovative, long-lasting, and well-designed products. The goal is to create a solution that combines functionality and their distinctive design in a way that blends in seamlessly with their product line.

The development will concentrate on the product's design, usability, and effectiveness to create a cleaning robot that reflects Husqvarna's high quality.

The proximity of Jönköping University to Torsvik, a major logistics hub with numerous warehouses, is a significant benefit of this project. Working with local businesses is made easier, and actual needs are better understood. Torsvik warehouses are crucial to supply chains and require routine cleaning to function properly. The project intends to address real issues and demonstrate Husqvarna's clever and dependable solutions by creating a robotic vacuum cleaner for this setting.

1.4 Purpose and Research Questions

To accomplish the goal of designing an autonomous cleaning robot intended for warehouse spaces, targeted towards Husqvarna, these research questions are asked:

[1] What design principles define Husqvarna's product identity?
[2] How could a conceptual design for a professional robotic vacuum cleaner look and function if Husqvarna were to create one?

1.5 Limitations

This project is a concept study aimed at exploring new ideas and forms rather than developing a product that is ready for production. The work will not cover technical aspects such as exact dimensions, material selection, or manufacturing. All the functionality of the cleaner will be presented at a conceptual level and not through technical explanations. The final prototype will only serve as a physical representation of the form and design language rather than a functional model. This work is also limited to the cleaner itself, excluding the design of a charging or docking station.

1.6 Disposition

The report is structured in a way that allows the reader to first gain an understanding of the project in the Theoretical Framework chapter. In the Methodology section, the reader will understand which methods were used in the report to achieve the desired goal. The Approach & Implementation chapter explains how these methods were applied and what needed to be done to achieve the intended outcome. The Results chapter presents the outcome through images and descriptions. Finally, the Conclusion and Discussion section reflects the research questions, highlighting what went well and what challenges arose during the project. An evaluation is also conducted of aspects that may require further development.

2. Theoretical Framework

2.1 What is product development

The process of developing products with new or changed features that provide customers with more or better value is referred to as product development. To satisfy a new consumer need or fill a specific market gap, this could involve improving an already-existing product, changing its appearance or design, or creating a completely new one [3].

2.2 What is design

Design is a word that can mean many different things. It can be about how cars, clothing, buildings, or even a website looks. Design can also be about how a service works. One example of this is how the steps in medical treatment are planned.

When it comes to people, there are two main goals with design:

- To make things comfortable and easy to use (this is called good ergonomics).
- To make things attractive and interesting for the people who will use them.

Within a company, design work should follow a clear plan that supports the company's goals and considers practical factors such as production, marketing, usage and recycling. It is also important to listen to feedback from customers and others involved, both inside and outside the company. This feedback helps create better and more successful designs [4].

2.2.1 Industrial design

Designing products that are used every day is the profession of an industrial designer. The focus areas for an industrial designer are appearance, functionality, and manufacturability for physical products. Designers are often involved in more than just creating the product's appearance. For example, they may take part in user testing, technical discussions, or marketing decisions during the development process. The objective usually starts with an assessment of the problem. Requirements like ergonomic functions, looks, and estimated sales volume are documented. The client will evaluate and approve the process to move forward.

Concept sketches are made based on information gathered by interactions with the target group and other stakeholders. Multiple concepts are compiled into presentations for the client to evaluate and confirm. The principal structure is created together with a prototype for testing. Detail design can now be implemented to create an ideal design. After the final prototype has been created, the design basis will be shared with a structural engineer for the next step of the process. The product is then followed up from a design point of view, for example the correct colour or finish [5] [6].

2.3 The design process

2.3.1 Design Methods

There's no single right way to approach design. Unlike the scientific method, which is often shown as a simple line, design comes in many forms, spirals, squares, double diamonds, and more. This variety shows that design is flexible and adaptable.

Different design methods help structure and guide the process while allowing room for creativity. Since design evolves through challenges, it improves with each iteration.

The Double Diamond represents a design approach focused on exploration, testing, and refinement. Its flexibility helps designers adapt to challenges and develop better solutions [7].

2.3.2 The double diamond process

The Double Diamond is a framework and visual aid that breaks down the design process into four stages: discover, define, develop, and deliver. Understanding the issue, determining user needs, and assessing how the issue impacts them are the main objectives of the first phase, Discover. Surveys and interviews are useful tools for learning about users.

In the second phase, define, insights gathered in the first phase are used to clearly define the core challenge.

The third phase, develop involves generating multiple solutions based on a well-defined problem, drawing inspiration from various sources, and collaborating with different people.

The final phase, Deliver, consists of testing and evaluating different solutions to identify the most effective ones, which are further refined and improved [8].

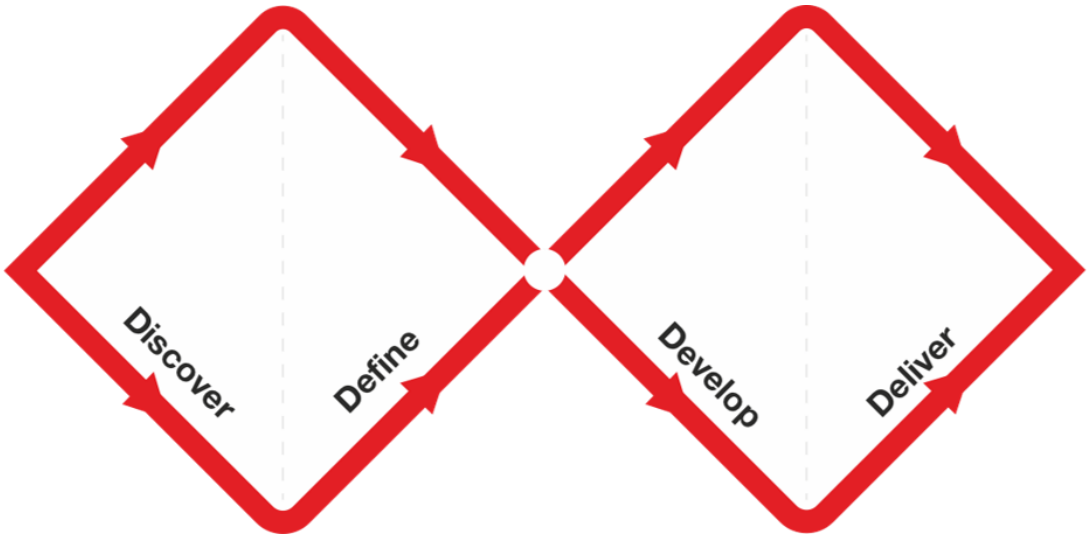


Figure 1: The double diamond process [8]

2.4 Form Language

Form language refers to the visual and aesthetic features of a product, such as shape, colour, material, and texture, that together create a recognizable style and communicate the product's identity and purpose. It plays a key role in shaping how a product is

perceived and understood by users, supporting both functionality and emotional response.

Form language is closely related to several established design concepts:

- Functionalism, where form follows function
- Emotional design, where aesthetic choices evoke specific user feelings

Husqvarna is a real-world example of form language development. Over time, their chainsaw design has changed to convey power and performance visually as well as to enhance functionality. This change in design, as mentioned in the thesis, is intended to appeal to new user groups and strengthen brand identity through visual expression.

By ensuring that the final product is both expressive and functional, these guidelines support the product's intended use and feeling through logical visual design [9].

2.5 Vacuum cleaner

By sucking up dirt, dust, and debris through an intake nozzle and collecting it in a collection chamber, a vacuum cleaner is an electrical device that cleans floors, rugs, and other surfaces.

An electric motor powers a fan, creating lower pressure inside the appliance, which causes higher-pressure air to rush through its nozzle. Finer particles are filtered out by filters as the air and the particles it contains pass through the cleaner, while heavier debris is collected in a dust bag or bin. While the gathered dirt is kept for eventual disposal, the cleaned air is then released back into the room [10].

2.6 Robot floor cleaner

A robot floor cleaner is an automated device designed to clean floors with minimal effort, reducing the need for manual cleaning. This conserves energy in addition to time. Modern robots can move through a room with ease, automatically dump the dirt they have accumulated into a container and then return to their charging station to refuel [11].

2.6.1 History

Since the first patents in the 1950s, robotic floor cleaners have gone through significant development. The industry began when Electrolux introduced the first model that was sold commercially in 1996. Later, iRobot's Roomba appeared as the first significant commercial success. Features like smartphone control, obstacle avoidance, and combined vacuuming and mopping have been made possible by developments in AI and sensor technology over time. A model with a mechanical arm that could lift small objects was revealed by Roborock in 2025. For large spaces like factories and warehouses, industrial cleaning equipment has advanced from manual tools to automated machines [12] [13] [14] [15].

2.6.2 Navigation Technology

The technology for automotive robots has for a long time have been based on a wire system that guides the robot within an area that is manually created. This process is labour-intensive and time-consuming.

To navigate without the need for physical boundaries, modern wireless autonomous robotic systems rely on advanced sensor technologies. Together, LiDAR, ultrasonic sensors, and cameras enable the robot to sense and navigate its environment.

By measuring the distances to nearby objects using laser pulses, LiDAR (Light Detection and Ranging) enables the robot to produce detailed three-dimensional maps of its surroundings. It helps the system in exactly planning safe routes and avoiding obstacles [16].

When high-frequency sound waves are emitted and bounce off objects, ultrasonic sensors measure how long it takes for the echo to return. This technique is helpful for navigating in confined or complicated areas because it is particularly good at detecting objects up close [17].

Cameras record visual data from the robot's environment. Real-time analysis of this data can be used to track movement, identify landmarks, and help with navigation, particularly in situations where GPS is not trustworthy or unavailable. When paired with other sensors, cameras are especially useful for enhancing the robot's perception of its surroundings [18].

These technologies work together to enable autonomous robots to locate themselves, understand their environment, and create effective routes.

2.7 Robot Floor Cleaner Considerations

Understanding important aspects such as the kind of cleaning that is required and the capabilities that are needed is crucial when designing a robot floor cleaner.

2.7.1 Floor cleaning process in warehouse

A clean floor is a vital component of a productive workplace. Dust and other debris are likely to get on warehouse floors, particularly if the space is used for product manufacturing or the movement of stored goods. The warehouse floor can be cleaned in a variety of ways. The procedure looks different depending on the kind of flooring and debris that is found. To get the best cleaning results, there are five general steps to follow.

1. Remove any objects or barriers that might be preventing the process from going more smoothly.
2. Any dirt or debris is eliminated by sweeping or vacuuming the floor.
3. Hard stains can be removed by scrubbing the floor. Following the advice given for the particular type of flooring is crucial.
4. To get rid of any last bits of dirt, rinse the floor with fresh water.
5. It is crucial to dry the floor to prevent it from becoming slick, as a slippery floor could become a risk for pedestrians and vehicles.

Frequent cleaning of the warehouse floor will greatly enhance the working environment [19].

The preferred way to remove dirt from the warehouse floor is to use a machine with a vacuum function. The vacuum will gather the dust instead of stirring it up into the air. For warehouses with areas of over 10,000 square meters, machines with a combination of functions such as Sweeping, scrubbing and vacuuming are recommended. Machines with side brushes are especially effective against removing loose dirt from edges and walls [20].

2.7.2 Robot Size

The market offers different sizes of warehouse cleaning robots for different uses and spaces. Smaller robots are more common in smaller warehouses and focus on collecting dirt and dust. The average range of a smaller robot cleaner is about 500 square meters.

Medium-sized cleaning robots are generally designed to perform a single task: collecting dirt and dust with the help of brushes and a vacuum system. These robots are efficient and can cover areas of up to 14,000 square meters

For the larger cleaning machines, the range gets up to 110,000 square meters before charge is needed. These machines are made for the largest warehouses and usually have all the necessary functions to maximize cleaning efficiency. These functions are sweepers, vacuum, and dryers [21] [22] [23].

3. Method

The Double Diamond process was used to carry out the project, and the following points describe the steps taken to achieve the results. The book Product Design and Development has been used as a foundation for the methods applied [24].

3.1 Identifying customer needs

The result of customer needs is used to guide the team in establishing product specifications, generating product concepts, and selecting a product concept for further development. The process of identifying customer needs include five steps [25].

1. Gathering raw data from customers.
2. Interpret the raw data
3. Organize the needs
4. Establishing the importance of the needs
5. Reflect on the result and the process

Identifying customer belongs to the first stage of the Double Dimond process, known as the “Discover” phase.

3.1.1 Market research

The process of market research is essential for gathering and analysing information about consumers, competitors, and market trends. It helps to make educated decisions regarding product development. There are different types of methods used to conduct market research. Primary research methods involve gathering firsthand data directly from the source. Surveys, interviews, and observation of consumer behaviour are examples of primary research. Secondary research methods involve analysing existing data from different sources, for instance reports and company websites.

3.1.1.1 Product board

A Product Board is a visual tool used to analyse a company’s design language by collecting images of its existing products. This helps identify key shapes and design elements that define the brand’s identity. By studying these features, a Product Board provides valuable insights that guide the development of new designs, ensuring they align with the company’s established aesthetic.

3.1.1.2 Trend analysis

To determine what is currently popular, trend analysis involves looking at current trends. Identifying frequently used materials, technologies, and colour schemes is part of this. To find more inspiration, it can be beneficial to explore more general trends in other industries, even though it is important to concentrate on trends within the relevant market.

3.1.2 Target audience analysis

Understanding the target market is important when developing a product. Target audience analysis is the process of identifying, learning, and evaluating the target market for a product or service. To improve a product's functionality and design, it is critical to understand its users. This is done through an understanding of important audience characteristics through focus groups, interviews, and final customer research.

3.1.2.1 Interview

To learn more about the target audience, interviews are important. Gaining a deeper understanding of user needs and experiences is the goal.

There are various formats for interviews. Semi-structured formats combine prepared questions with spontaneous questions based on participants' responses. When trying to gather both specific data and deeper insights, a semi-structured approach is often recommended.

Depending on the circumstances and requirements of the study, interviews can be done online or in person and typically last 45 minutes to an hour. To record important details, notes are typically taken during the interview.

3.1.2.2 Persona

Personas are fictional representations of typical consumers of a product or service. They are used to better understand the needs, difficulties, goals and experiences of the user. Personas help guide the design process by ensuring that solutions are based on real user needs, as identified through previous research. By focusing on these fictional users, it becomes easier to identify potential problems and develop a product that improves overall satisfaction and usability.

3.1.2.3 Mood board

One visual tool for defining the general style, feeling, and direction of a design project is a mood board. By combining textures, colours, materials, and images that reflect the desired look, it helps create a unique visual theme. Throughout the design process, mood boards offer consistency and serve as a source of inspiration. They help by turning general ideas into an original design that supports the project's objectives.

3.2 Product specifications

To guide the design and engineering of a product, the team defines a set of specifications, outlining what the product must achieve to be successful. These specifications should be based on customer needs and should also distinguish the product from its competitors [26].

Defining product specifications takes place in the second stage of the Double Diamond process, known as the "Define" phase.

3.2.1 Product MoSCoW

MoSCoW is a method used to help prioritize different product features into four groups. These are Must have, Could have, Should have, and Won't have. By structuring the features, it is easier to know what to focus on to create the desired product.

Must have: Features in this group are critical for the product's success. If one of the features is not included in the final product, the project is considered a failure.

Should have: Features in this group are important but not vital for the product's success.

Could have: Features in this group are desired but not necessary for the product. These will be included if the project allows extra time and cost.

Won't have: Features in this group are the least critical for the product but may be added if the project allows the time and money in further versions at a later time.

3.3 Concept generation

When developing product ideas, it helps to follow guidelines to boost creativity and get better results. Brainstorming is a good way to come up with many ideas without judging them too early. By keeping an open mind, you have a better chance of finding useful solutions. The more ideas you generate, the better. Strange or weak ideas can also lead to great results. It is important to approach the process with creativity and an open mind, as even unconventional ideas can often be developed into effective solutions. Quick sketches help make ideas easier to understand and develop [27].

Idea generation occurs in the third stage of the Double Diamond process, called the "Develop" phase.

3.3.1 Brainstorming

A creative method used in groups to generate concepts and ideas for solving one or more problems is brainstorming. To generate a wide range of ideas for future development, this method is useful at the start of a product development process. By writing or sketching ideas, it is possible to quickly visualize different solutions for form or function. To further develop the design without restricting the creative process, three key guidelines should be considered. First, it is important to focus on a large number of ideas, placing quantity before quality in the early stages. Second, there should be no limitations. All ideas are welcome. Lastly, criticism should be set aside, as even an "impossible" idea can serve as inspiration for a new and improved concept.

3.4 Concept selection

Concept selection is important because it helps find which product ideas are most likely to succeed. It makes sure that the selected ideas match customer needs, enabling more informed decision-making. This helps prevent time and effort from being spent on ideas that are unlikely to succeed. A good selection process also makes it easier to explain and support the final choice.

An external decision occurs when the responsibility for selecting a concept is handed over to a client, customer or another external party, for example, when the product is being developed on behalf of a company that will ultimately manufacture or market it.

Intuition-based concept selection relies on a gut feeling or instinct to choose a concept. No formal criteria or trade-offs are considered. This can be done if the concept simply feels like the better choice [28].

Concept selection is also an essential part of the "Develop" phase in the Double Diamond process.

3.5 Concept testing

To get direct feedback on a product idea from potential customers, concept testing is used. It helps to decide whether the concept satisfies consumer demands and has a good chance of succeeding in the market. Concept testing looks at how customers respond to the idea. It can be helpful in different phases of product development, especially when choosing which concept to pursue. Typically, the process includes creating a goal for the test to receive relevant feedback. An example of this can be evaluating colour, dimensions, and form. Computer-aided design, renders, and prototyping helps explain the idea, which can be useful to get important information for further development [29].

Testing is a key component of the final stage in the Double Diamond process, known as the "Deliver" phase.

3.5.1 Computer-aided design

Starting with an initial idea or sketch, Computer-Aided Design (CAD) serves as an effective and flexible tool for developing detailed product designs. It makes it possible to create accurate three-dimensional models with exact dimensions, proportions, and features. The process of making changes is one of CAD's main benefits. Designs concepts can quickly be tested, evaluated and refined without the need to start from scratch.

In addition to better technical accuracy, CAD helps communication during the design process, making it simpler to share and analyse designs with others involved.

3.5.2 Render

With pictures or animations, renderings are used to show the product's materials, colours, shape, and functionality. Renderings can be made from an existing 3D model to provide detailed, realistic images that are often easier to make than real photography.

Many companies choose to use rendered products in their advertising. One reason is that it is cheaper than doing real photo shoots. With 3D renders, companies have full control over lighting, angles, and background.

3.5.3 Prototyping

Creating a physical prototype can help visualize the final appearance of the product by following these steps:

1. 3D-print the components of the base form
2. Use coarse sandpaper to achieve the desired surface and correct visible imperfections.
3. Apply spackle to smooth uneven areas and fill gaps.
4. Gradually use finer sandpaper to further refine and even out the surfaces.
5. Apply primer to the parts.
6. Perform fine finishing sanding.
7. Paint the parts in layers until desired effect.
8. Assemble the finished components into a complete prototype.

4. Approach & Implementation

4.1 Product Board

To guarantee that the new design reflects Husqvarna's unique design identity, a product board was created. By collecting and analysing existing Husqvarna products, key shapes, design elements and form language were identified. This analysis serves as a foundation for understanding the visual characteristics that define the brand. The analysis shows that Husqvarna's products are based on the principles of durability, functionality and innovation. The design combines soft and sharp transitions as well as distinct contours. The insights gained from this process help guide the development of a new design that remains true to Husqvarna's aesthetic. The following form analysis (*Figure 2*) highlights the key design principles identified.

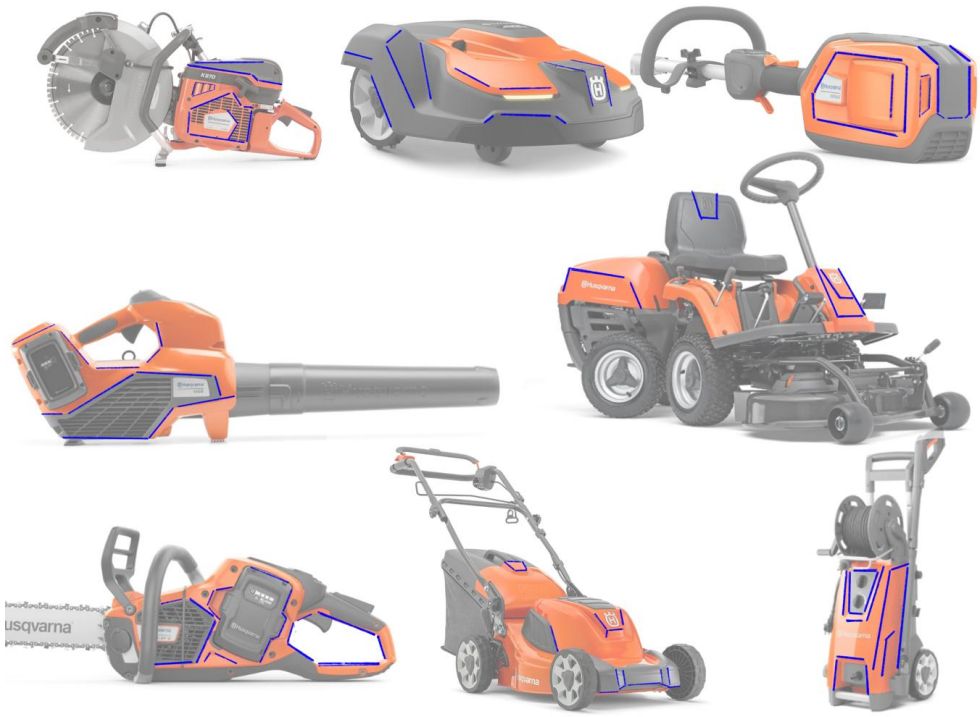


Figure 2: Product board

4.2 Trend analysis

The analysis is based on trends in warehouse environment, automation, and existing products that address similar problems today. Robots that clean warehouses are made to be efficient and long-lasting, with a modern yet practical exterior that can survive challenging industrial environments. They are optimal for high-traffic areas because of their sturdy design, which guarantees a long lifespan. Reinforced materials shield against impact, moisture, and dust. The use of advanced sensor technology is an important design component that affects the robot's overall form. Even though they can perform heavy tasks, many of these robots are made to be small and nimble so they can move through tight aisles and small warehouse areas.

4.3 Interview

The interviews took place at an IKEA distribution centre in Torsvik, which is located just outside of Jönköping. To learn more about their everyday struggles with warehouse cleaning, three Samhall employees were interviewed.

Samhall assists IKEA by managing cleaning tasks, such as keeping warehouse floors clean. The interviews took place at the warehouse and lasted for about an hour. Multiple issues were explored using a semi-structured interview format that allowed open-ended responses and spontaneous follow-up questions. Most questions focused on the cleaning process and potential areas for improvement.

The interviews highlighted several challenges in the current cleaning operations at IKEA's distribution centre in Torsvik, where Samhall employees are responsible for the work. Although a versatile scrubbing machine (Scrubmaster 400B RH, 2022) is used daily, many difficulties remain.

Cleaning under shelves and conveyor belts, where dirt may gather easily, are the primary issues. Workers must manually clean these difficult-to-reach areas, which results in uncomfortable working conditions and physical strain. Debris frequently blocks the scrubber machine, needing the operator to stop and manually fix it several times a day. The information gathered where noted for later importance in the project.

- Poor accessibility under shelving and conveyor belts
- Frequent machine blockages requiring manual handling
- Ergonomically challenging work conditions
- Lack of efficient solutions for collecting larger debris

The employees suggested smarter warehouse layouts that make cleaning safer and easier, more automation, and better-designed machines as ways to improve the process.

4.4 Personas

Based on information gathered through employee interviews, two personas were created to better understand the needs, challenges, and priorities of potential users. These personas, which reflect their daily routines, problems, and objectives, represent the product's target market.



Name: Anna Eriksson
Age: 32
Occupation: Warehouse Cleaner
Location: Jönköping, Sweden

General information

Anna Eriksson has been working as a warehouse cleaner for the past six years. She enjoys her job and takes pride in maintaining a clean and safe working environment for her colleagues. She values stability and appreciates the structured routine her work provides.

Pain points

- Started experiencing back pain due to the physically demanding manual tasks in uncomfortable positions
- Often feels that her time could be better utilized if some of the more repetitive or physically demanding tasks were automated or made easier

Figure 3: Persona - warehouse cleaner



Name: Gunn Söderberg
Age: 56
Occupation: Warehouse Manager
Location: Värnamo, Sweden

General information

Gunn Söderberg has over 25 years of experience in warehouse operations, with 10 years as a manager. She is dedicated to efficiency, worker well-being, and continuous improvement. She values teamwork and seeks innovative solutions to optimize workflows while reducing costs.

Goals & Priorities

- Minimize physical strain with ergonomic solutions
- Reduce costs while maintaining efficiency, quality, and employee satisfaction.

Figure 4: Persona - warehouse manager

4.5 Mood board

To identify elements and expressions that reflect both the future and a strong design identity, this mood board was created, drawing inspiration from Husqvarna's existing products. The collage features a mix of products and design details that align with a robust and modern design language. The recurrent orange colour serves as inspiration during the ideation stage and supports an industrial and functional impression. The mood board also communicates a sense of robustness and durability, which is crucial for a product made to survive the harsh conditions found in warehouse settings. It communicates a balance between durability, functionality, and innovation.



Figure 5: Moodboard

4.6 MoSCoW

Insights from the conducted interview combined with a market analysis, helped identifying key features that were used to define requirements when generating ideas. Features were categorized into four groups depending on the importance. requirements in the group “Must have” are vital for the project’s success.

Must have	**Should have**
Clean dirt and dust from the warehouse floor Autonomous cleaning and spatial awareness Durable and robust for warehouse environments Safe for surroundings and workers User-friendly stop function	Automatic waste disposal Easy maintenance and service Customizable routes and cleaning schedule Connectivity for remote monitoring and steering Dust filtration system Automatic charges via docking station
Could have	**Won’t have**
Collaboration between multiple units Self-repair diagnostics Features for moving objects AI-Trash recognition Water-based cleaning	High-speed movement Climbing stairs

Table 1: MoSCoW

4.7 Brainstorming

To generate and explore a wide range of ideas that could later be sorted and refined, the first step in developing the concepts was to quickly draw sketches. At this point, the quantity and creativity of the ideas were more important than the quality. Rapid sketches helped with idea sharing and discussion, enabling the team to consider a wide number of alternatives.

The process began with individual sketching sessions, where each participant generated ideas independently. These ideas were then compared and discussed. The process was repeated once more, allowing participants to build on previous ideas or explore new directions. The resulting sketches were sorted, categorized, then presented and discussed with the supervisor from SEMCON. Together with their feedback, a selection process was conducted to eliminate less viable options and identify the ideas with the greatest potential for further development.

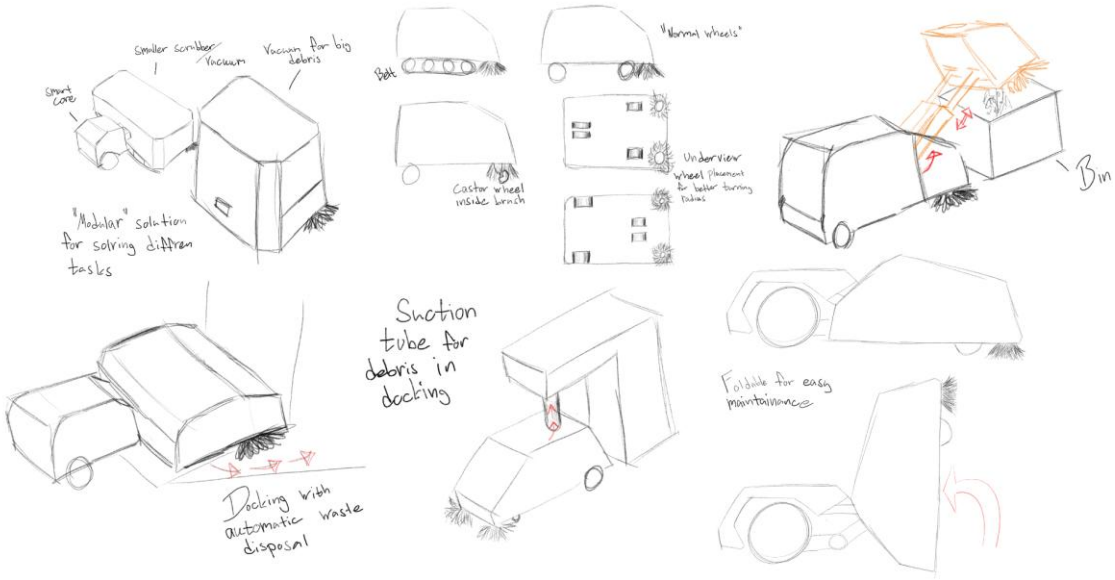


Figure 6: Brainstorming sketches

4.8 Concept ideation

Using ideas generated during the brainstorming phase, the concept ideation stage began with the creation of various shapes and forms. A broad range of design explorations made it possible to identify and refine preferred aesthetics. By developing several distinct concept directions, the process naturally gravitated toward a desired design path.

At this stage, much inspiration was drawn from Husqvarna's existing product range. While Husqvarna had no current offering in the warehouse cleaning robot segment, the closest reference point was their line of robotic lawn mowers.

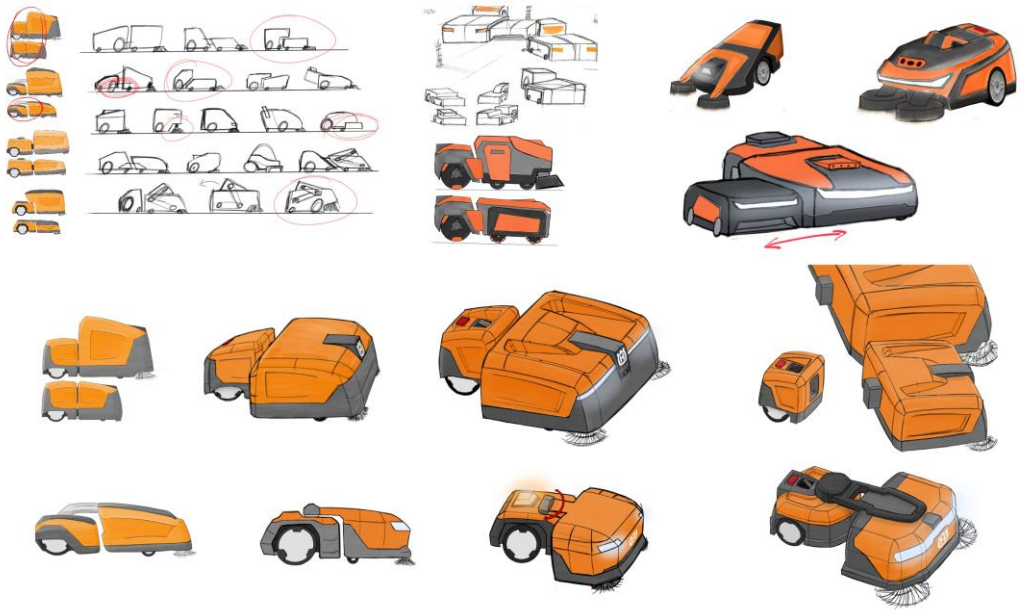


Figure 7: Ideation sketches

4.9 Mid-presentation design concepts

In the middle of the project, a presentation was held to showcase the five developed concepts. Each concept featured different functions and forms to help determine which direction should be pursued further. All concepts were carefully evaluated based on their innovative features, visual design, and how well they would fit into a warehouse environment. The concepts were evaluated by the project owners in collaboration with supervisors and professionals from SEMCON.

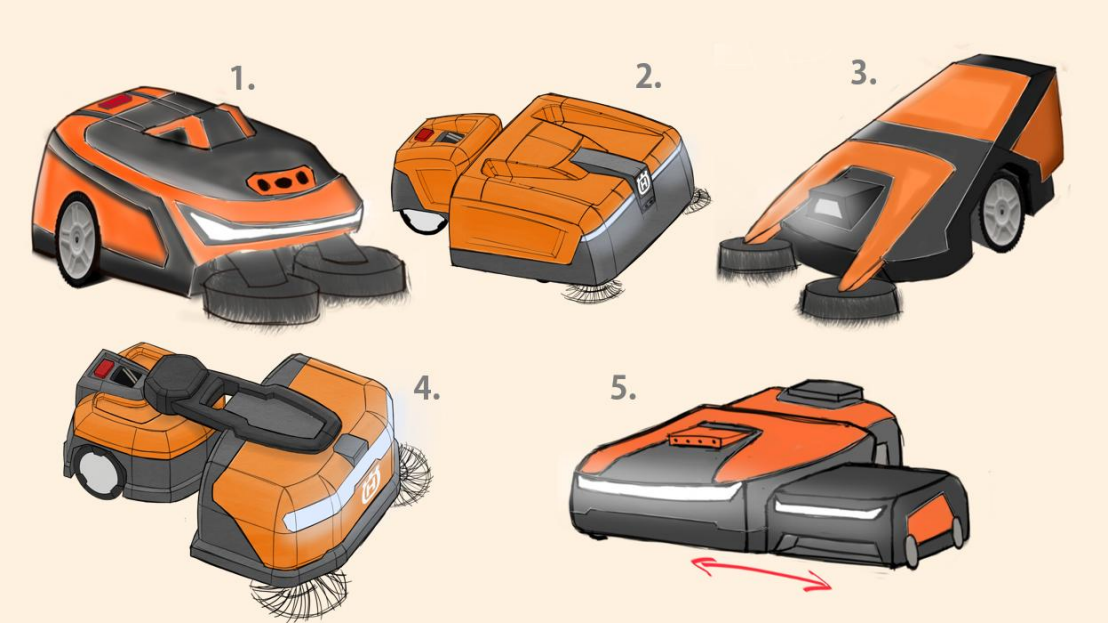


Figure 8: All concepts

4.9.1 Concept 1

The focus of this concept was the features of picking up larger litter with the help of two big brushes gathering all the trash in a container at the end of the robot. When the collecting space is full, the robot will automatically locate the nearest trash bin for disposal. This feature is based on two arms that move the container full of trash in a circular motion backward into a trash bin. The design is intended to communicate a sense of robustness, ensuring it aligns with the demands of a warehouse environment.

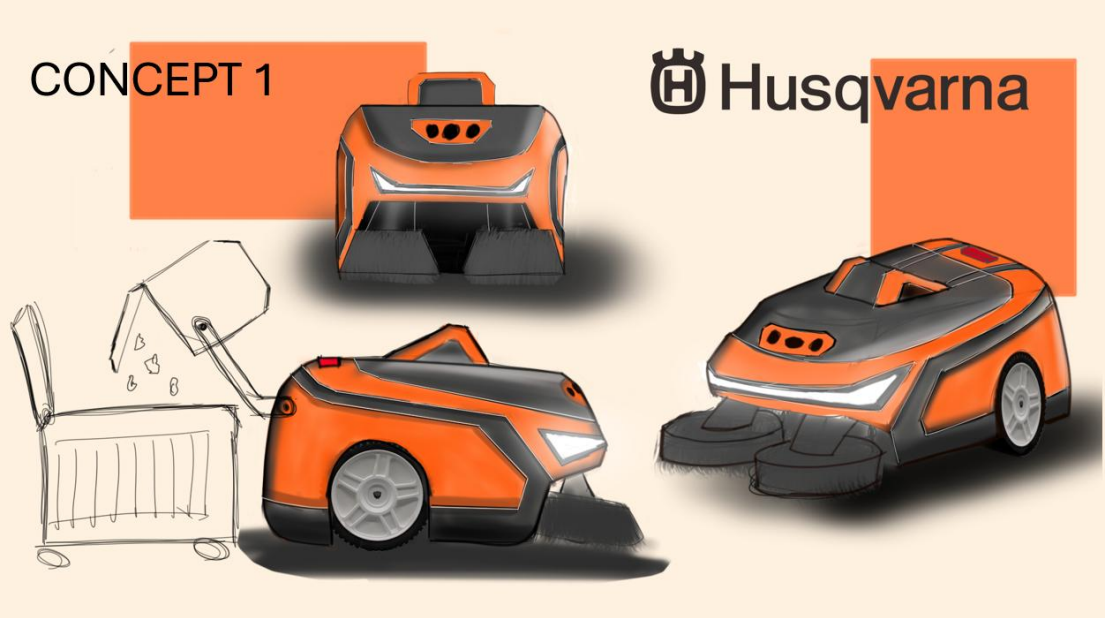


Figure 9: Concept 1

4.9.2 Concept 2

Concept 2 presents an innovative solution to the challenge of varied cleaning requirements within a warehouse environment. The concept uses a modular system to enable multiple specialized cleaning methods by breaking the robot down into several components. The system is made up of a small main unit that is connected to detachable front modules, each of which is made to perform a particular cleaning function.

The main unit has essential features like rear drive wheels, a control panel, and a safety emergency stop button. Regardless of which front module is connected, this unit is always in use and serves as the centre of power and control.

Cleaning is usually necessary in both open aisles and the tight areas under shelving units and conveyor belts in warehouses. This is solved by the design of one front module, which collects finer dirt and dust by reaching under low-clearance areas, and another that handles larger debris in the aisles. This modularity allows the robot to adapt effectively to different cleaning scenarios within the same environment.

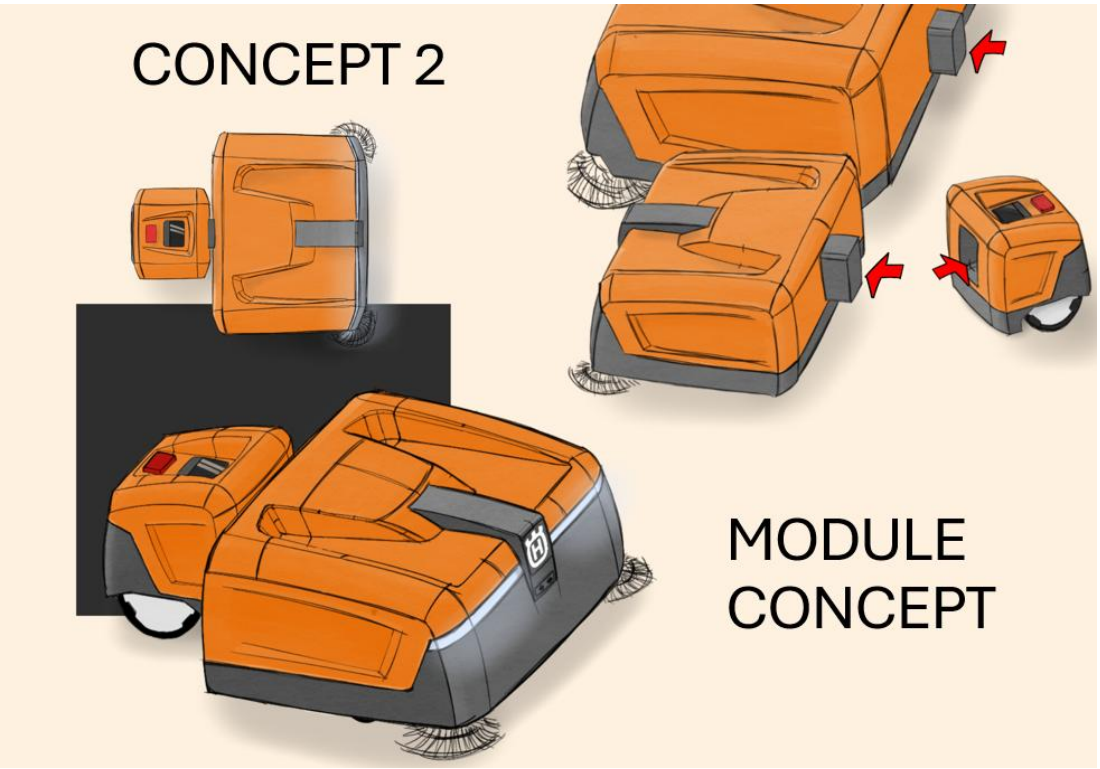


Figure 10: Concept 2

4.9.3 Concept 3

Similar to Concept 1, the goal of this concept is to gather larger debris that ends up in warehouse aisles. The garbage is swept by two brushes into a large storage space at the robot's back. The robot will navigate to a set point and wait for manual emptying once the compartment is full. Inspired by some of Husqvarna's manual lawn mowers, the design seeks to communicate a sense of durability.



Figure 11: Concept 3

4.9.4 Concept 4

In order to maximize trash collection, Concept 4 investigates the idea of a split robot, in which the front section can rotate on its own. The front can pivot side to side, allowing the brushes to reach and gather as much dust and debris as possible, while the fixed rear keeps the robot moving in a constant direction. In small areas with restricted human access, this feature works especially well.

A position light is installed above the headlights to improve visibility. Forklifts in warehouse settings frequently have this feature. A safer and more effective work environment is created by position lighting, which increases visibility in crowded areas and around corners.

As in concept 2, the rear part also has a control panel and a stop button for manually looking at technical features and stopping the robot in necessary situations.

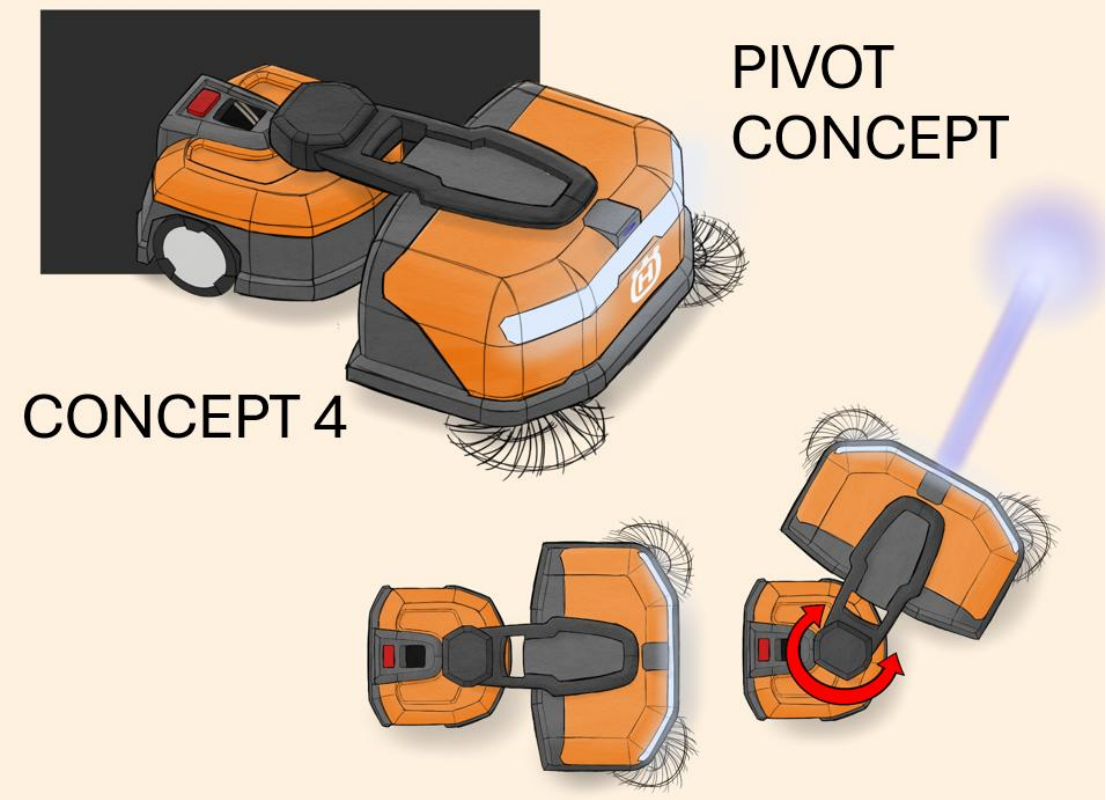


Figure 12: Concept 4

4.9.5 Concept 5

The fifth concept is built around a modular system, designed to offer maximum flexibility when cleaning warehouse floors. The concept consists of three main components. The rear section of the robot, although smaller in volume, functions as the "brain" of the system. This core unit can connect two different front modules, depending on the cleaning task at hand.

The smaller front module is designed to tackle dust and dirt in areas that are difficult for humans to reach, such as under conveyor belts or shelving units. Once aligned with an aisle, the robot extends horizontally to access these tight spaces. Omni-wheels are used to enable this smooth lateral movement.

For larger debris, the core unit can instead connect to a different module specifically designed for heavy-duty cleaning. This front module uses two large brushes to collect trash and can be manually emptied once the storage compartment is full.

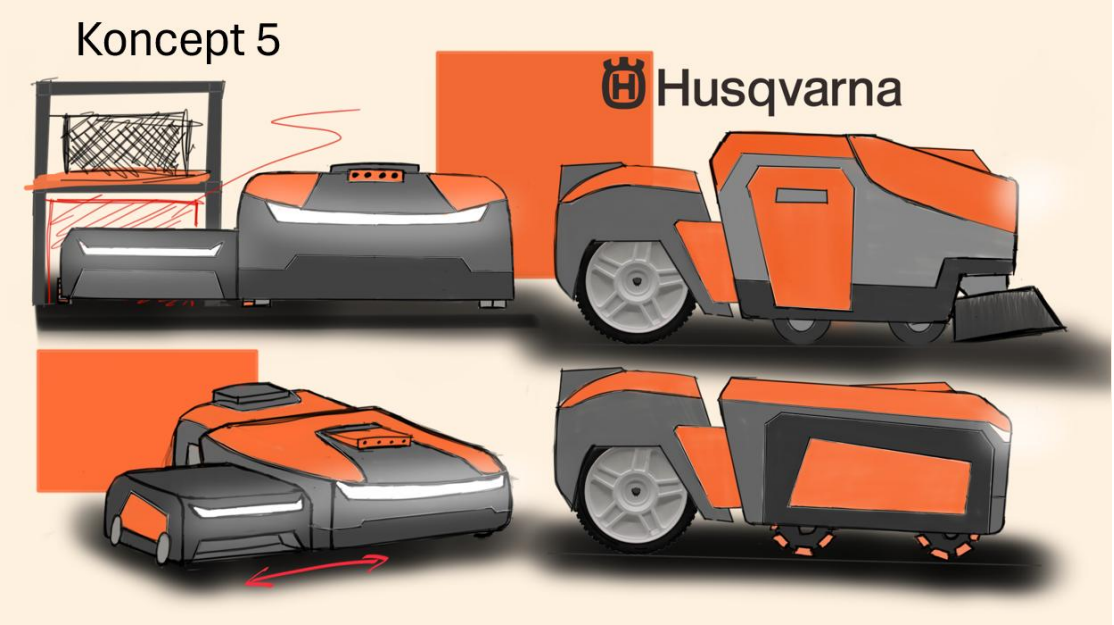


Figure 13: Concept 5

4.10 Design update

After the different concepts were evaluated during the mid-project presentation, Concept 2 was found to be the most creative and likely to advance. It provided a unique solution to the issue and the best representation of Husqvarna's brand identity. But more work was required to improve the design's functionality and aesthetics. Sketching out desired features and aesthetics, inspired by the presentations feedback, as well as experimenting with and modifying the concept's final dimensions, led to improvements.

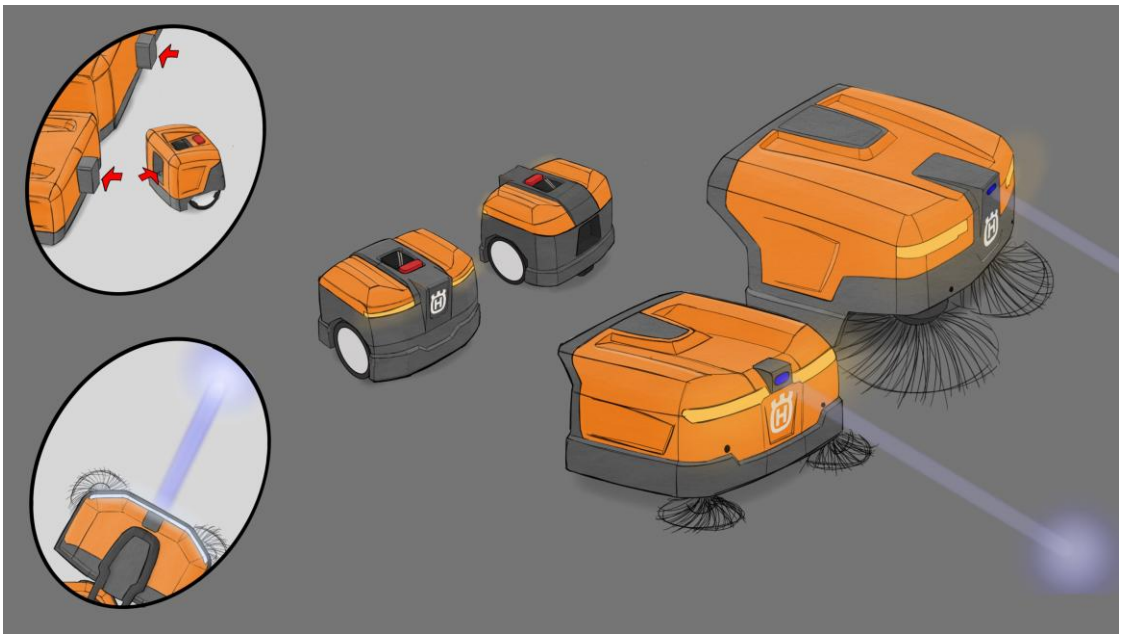


Figure 14: Updated design

4.11 Dimension testing

Two 3D models were made in SolidWorks using the updated design sketch as a guide. This allowed for a clearer visualization of the robot's dimensions, shape, and proportions. The work in the following stages could be completed more quickly and with fewer surprises after the digital model was complete.

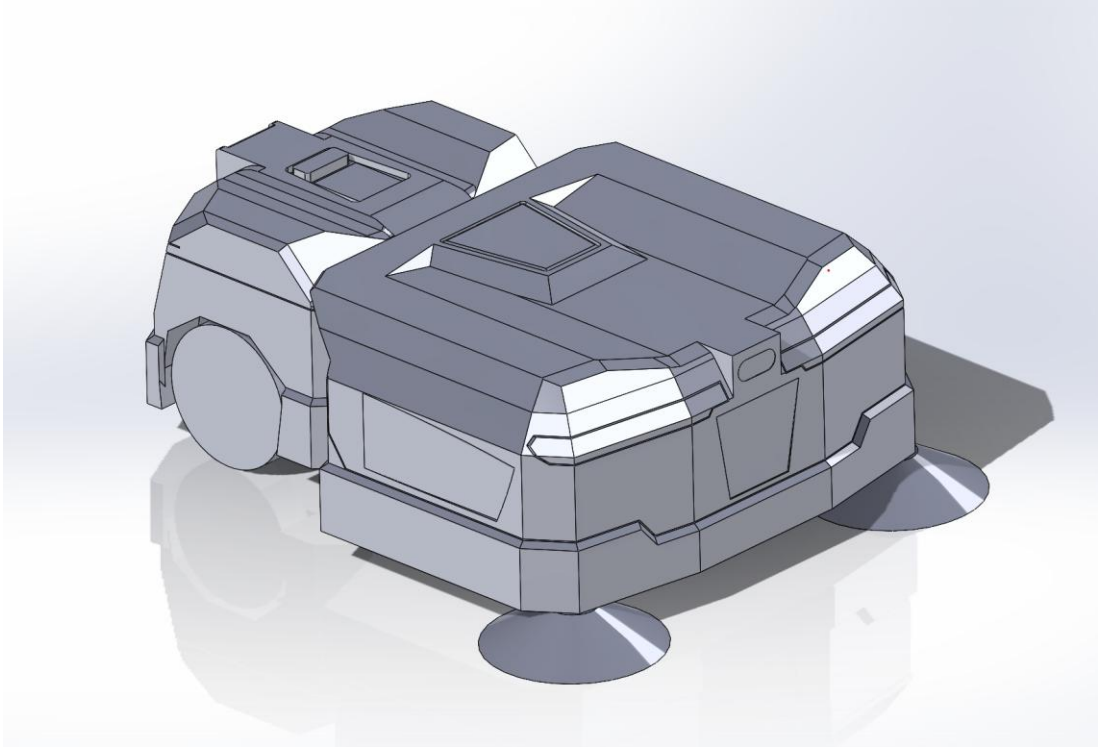


Figure 15: Solid model of the small front part in SolidWorks

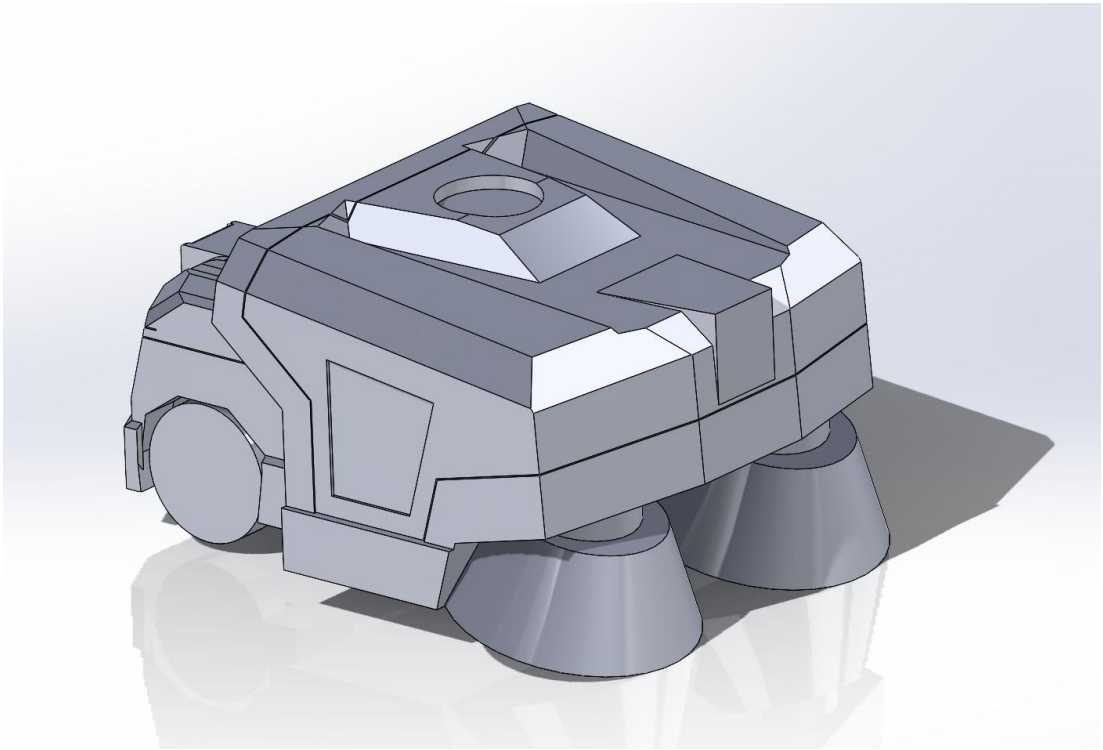


Figure 16: Solid model of the big front part in SolidWorks

4.12 Final design sketch

After defining all the necessary design elements, the final design sketch was created. Form, colour, and functionality choices were made with the project's primary goals in mind. An iterative process was used to create the sketch, with a focus on user-centred design and visual clarity. The result is a clear and thoughtful idea that communicates the project's goal and purpose.



Figure 17: Final design sketch

4.13 Surface modelling

Surface modelling was used to create curved surfaces matching the final design sketches. By creating curved lines, surfaces could be created to finally build a complete model of the concept. Generative Shape Design (GSD) and Part Design were used in the software CATIA to create the model. The software was used based on acquired experience close to the execution and after discussion with the supervisor from SEMCON.

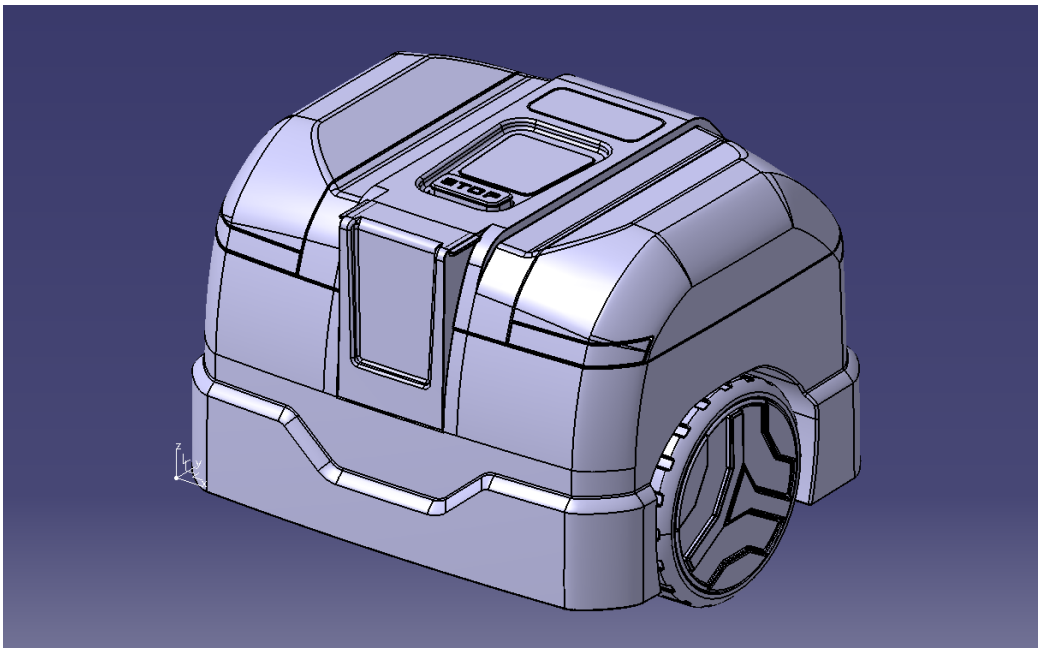


Figure 18: Surface model of the core part in CATIA

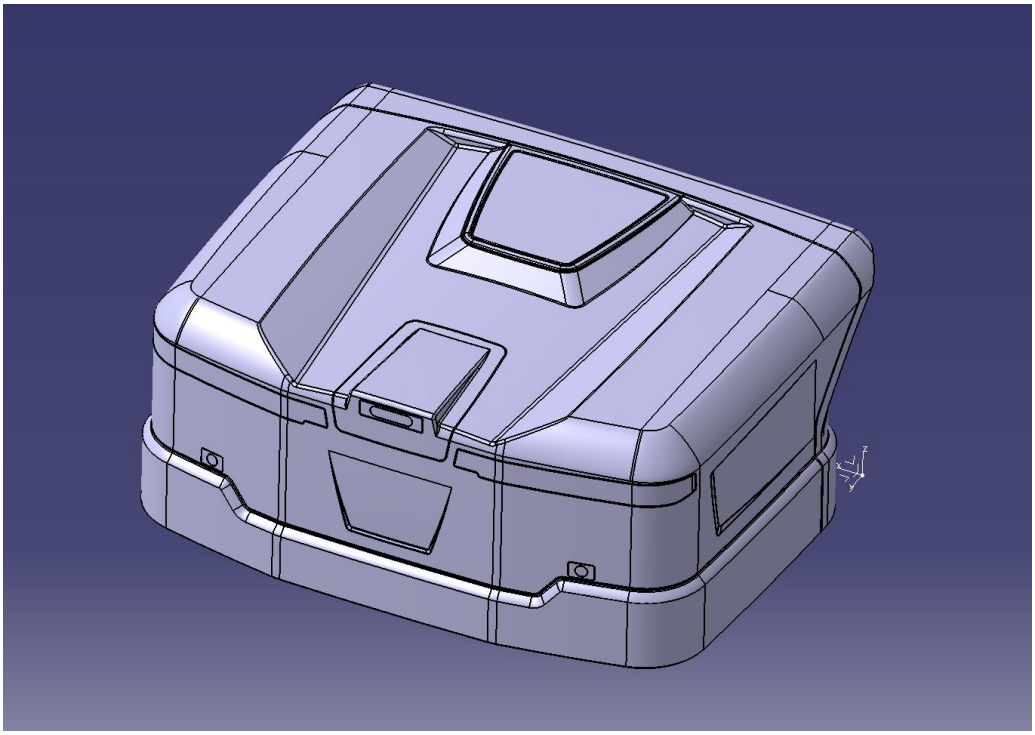


Figure 19: Surface model of the small front module in CATIA

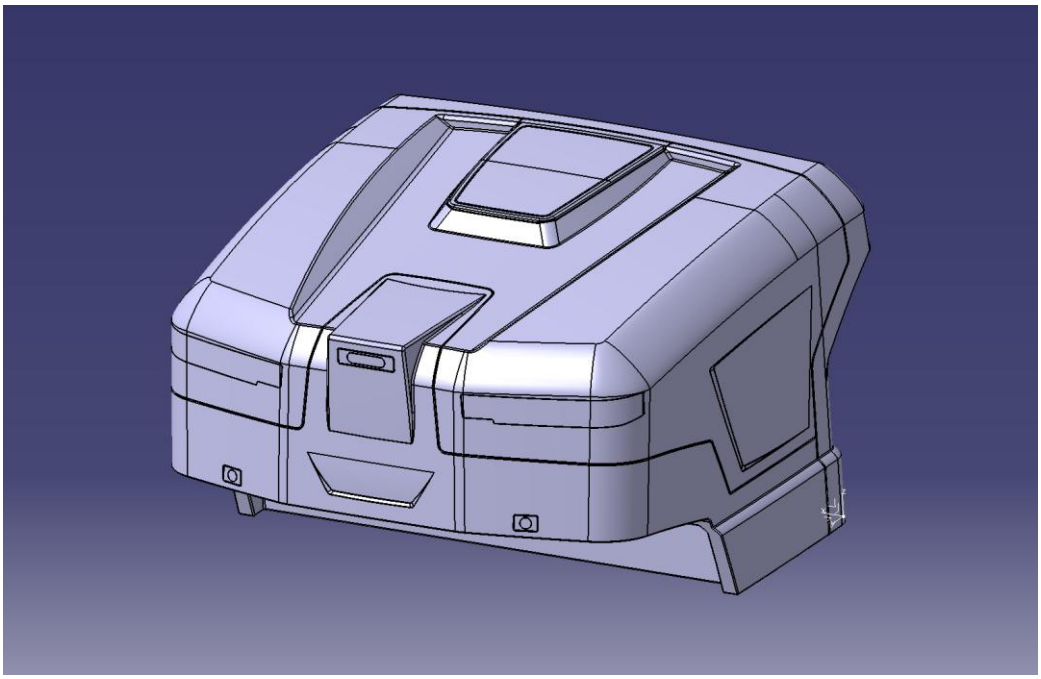


Figure 20: Surface model of the big front module in CATIA

4.14 Prototype

A prototype was developed to visualize the project's goal of designing a conceptual warehouse cleaning robot inspired by Husqvarna's design language. By strategically dividing the complete surface model, multiple parts could be 3D-printed and later assembled. PLA plastic was chosen for the prototype due to its accessibility at JTH.

Before the final assembly, each part went through a series of finishing steps including gluing, puttying, sanding, priming, painting, and a final round of gluing. A custom wheel axle was also created and installed, allowing the larger rear wheels to rotate freely. The prototype was constructed in facilities provided by JTH.

The prototype was mainly made to show a scaled version of the final design, both as a visual model and a display piece for an exhibition.



Figure 21: Prototype in parts

5. Result

5.1 Design

The result of the project is an autonomous robot for warehouse cleaning designed as if it was a product that the company Husqvarna would sell in the future. The purpose of the concept is to showcase innovative solutions for a product in an undiscovered market and at the same time express Husqvarna's form language. The design is meant to mediate the feeling of innovation, sustainability and functionality.

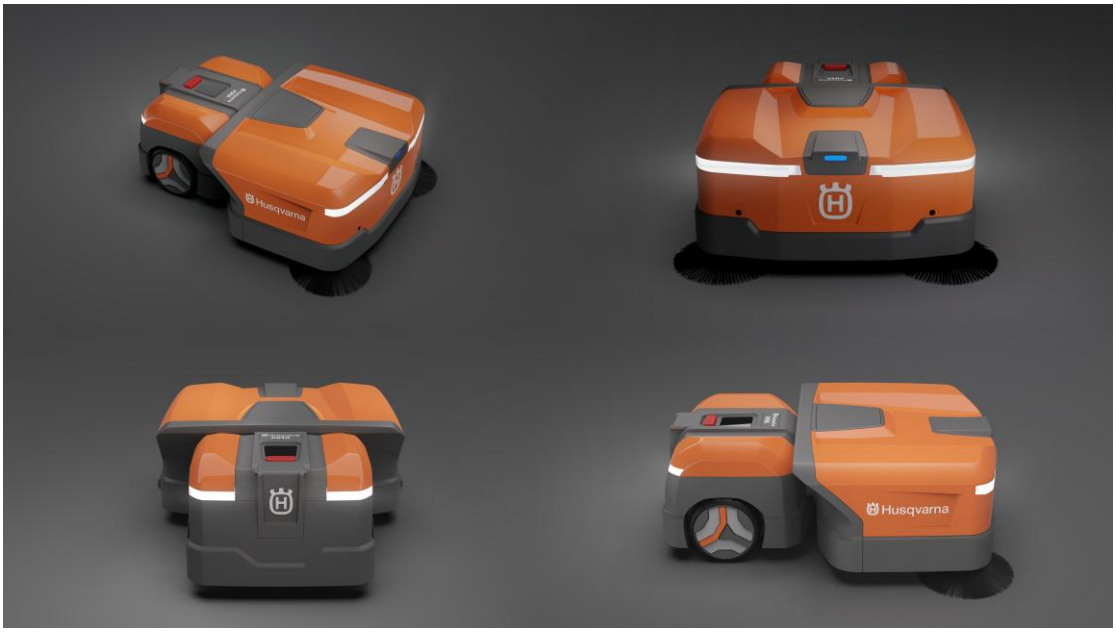


Figure 22: Design with small front module



Figure 23: Design with big front module

5.2 Product name

The product is called Pura to reflect the Latin meaning of the word, which is "pure" or "clean". The name was chosen to give a feeling of simplicity, quality and clarity, values that match the product's design and purpose. The idea for the name came from Husqvarna's product Ceora, which stands out with its unique and modern sound. Even though Ceora does not have a meaning in Latin, its name helps it stand out from other similar products. In the same way, Pura was chosen not only because of its meaning but also because it sounds clean, elegant and different. It helps the product feel unique while still being easy to remember.



Figure 24: Product name

5.3 Product features

5.3.1 Cleaning function

The concept's primary goal is to remove dirt and dust from warehouse floors. Two different front module types, each with specialized brushes and a vacuum system to help collect dust and debris, provide the cleaning functionality.

The dirt tucked around poles, corners, and other confined spaces is efficiently swept up by the smaller brushes in the module made for small, difficult-to-reach places as shown on the left side in *Figure 25*. Once gathered, the debris is vacuumed into an internal container within the robot.

The larger front module is equipped with two wide brushes intended for collecting larger debris commonly found in warehouse aisles. This dual-module system ensures efficient cleaning across a variety of environments and surfaces. These functions are directly inspired by insights from the warehouse interviews, aiming to address the issues of unergonomic work for the cleaning staff.



Figure 25: Comparison of the two modules

5.3.2 Modularity

The robot's modular design enables it to switch between various cleaning modules, each created to address distinct indoor cleaning issues. While one module is designed to handle larger debris typically found in open aisles and walkways, the other is optimized for reaching and cleaning under conveyor belts and in the small spaces between storage racks. Using a smart docking system, the robot automatically connects to the appropriate module, allowing quick and easy transitions without the need for human assistance. The need to buy and maintain separate machines for various cleaning jobs is eliminated with this modular approach. Instead, all cleaning functions are integrated into a single, adaptable product that provides full coverage of complex indoor environments while reducing both costs and storage requirements.

This also allows necessary maintenance of the modules without interrupting the work. If a module breaks or parts such as brushes need to be replaced or cleaned, the main unit can continue cleaning by using another one in the meantime.



Figure 26: Switching modules

5.3.3 Charging

Just like a forklift in a warehouse needs to be recharged, this robot will also need to recharge its battery. This is being done at a charging station placed in a location suitable for the warehouse environment. The autonomous robot is designed to reverse into the station to connect for charging.

At the rear of the main unit, there is a tower equipped with two charging pins that make direct contact with the station, allowing it to recharge efficiently. This type of charging is like how Husqvarna's robot lawn mowers dock today.



Figure 27: Charging

5.3.4 Trash disposal

Once the robot's internal trash container reaches full capacity, it needs to be emptied. This process takes place at a combined charging and disposal station. While the robot is being charged, it is simultaneously emptied via a large vacuum pipe connected to the front module of the robot.

When the robot is docked, the vacuum system is designed to automatically connect to either of the cleaning modules. The gathered debris is then removed by the vacuum pipe and moved to a bigger central waste storage unit. This setup guarantees a fully automated, effective maintenance process requiring little human involvement.

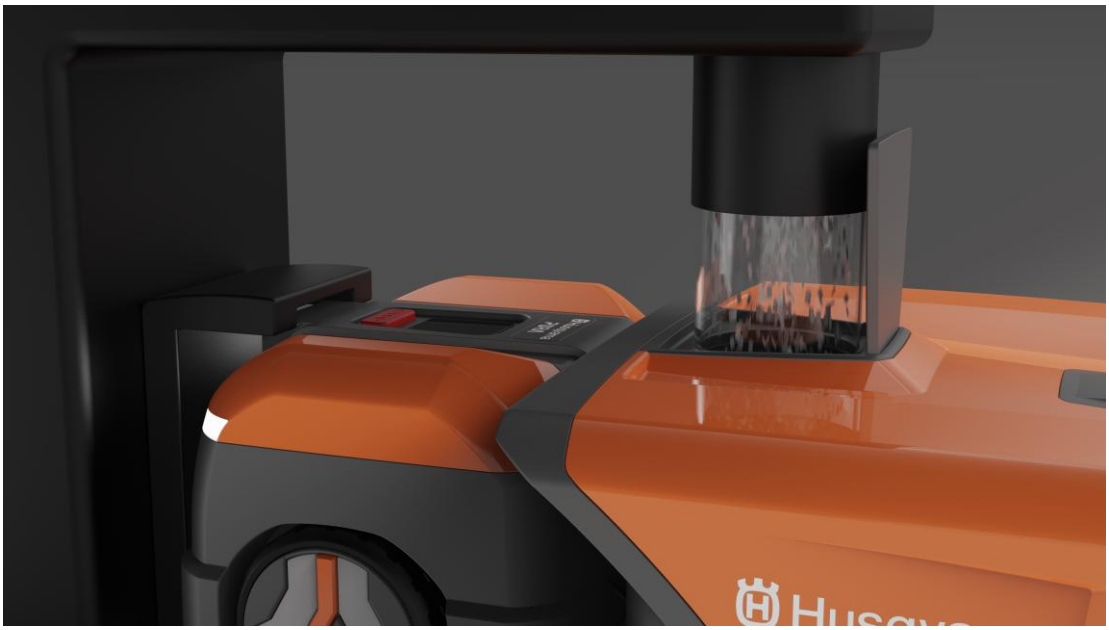


Figure 28: Autonomous trash disposal

5.3.5 Control panel

To enable the user to control and personalize the cleaning robot's activity, a control panel is used. The user can choose a particular area to be cleaned, select the type of cleaning that is required, set an active schedule, and select other important options through the control panel. Viewing the robot's status, keeping an eye on battery levels, checking for error messages for better service and maintenance, and getting real-time updates on the cleaning process are all additional important functions.



Figure 29: Control panel

5.3.6 Navigation

For the cleaning robot to operate effectively and safely in the warehouse setting, a navigation system is necessary. The robot can map its environment, identify obstacles, and create the best cleaning routes by combining sensors like LiDAR, ultrasonic sensors, and cameras. The robot can locate itself within the warehouse, avoid collisions, and adjust to changing environments with moving objects thanks to the navigation system. Additionally, it allows the robot to return to its charging station when the cleaning task is finished, or the battery is low. The user can set virtual boundaries, see the mapped area, and track the robot's location and cleaning progress in real time using the control panel.



Figure 30: Image sensor

5.3.7 Position light

A commonly used function in warehouses is the safety position light. It is usually mounted on forklifts to project a light a few meters ahead or behind, alerting people to the vehicle's approach. This feature improves workplace safety, especially in aisles with reduced visibility. Since the autonomous cleaner is smaller than conventional warehouse vehicles, it is even more important to include indicators that make its presence known.



Figure 31: Position light

5.4 Multicolour options

The dark grey and orange colour scheme that is used frequently in Husqvarna products today has grown to be an iconic characteristic of the brand. However, Husqvarna currently offers customization options for certain robot lawn mowers by providing a range of colour variations.

The concept that has been developed targets a market segment that is currently unexplored within Husqvarna's product range. This opens an opportunity to introduce new colour schemes that reflect the unique characteristics of this emerging market, helping the product stand out while still aligning with the overall brand identity. Several different colour variations were tested to visually demonstrate how the product could have looked.



Figure 32: Colour test

5.5 Prototype

A prototype was created using PLA plastic to provide a physical version of the product. The purpose of the prototype is to give users and viewers something real to hold and interact with. It serves as a visual and physical representation of the final design and is especially useful for showcasing the concept at exhibitions. While not functional, the prototype helps communicate the product's shape, proportions, and overall design more clearly than digital models alone.



Figure 33: Finished prototype

6. Conclusion and discussion

6.1 Research questions

This chapter will highlight the project's goals and assess whether those goals have been achieved. The research questions have guided the process of designing an entirely new product. They have served as a foundation for decision-making and helped ensure that the development stayed aligned with the user's needs and the project's overall vision.

[1] What design principles define Husqvarna's product identity?

[2] How could a conceptual design for a professional robotic vacuum cleaner look and function if Husqvarna were to create one?

6.1.1 Husqvarna's product identity, Research questions 1

To create a Husqvarna product, it is important to understand the design principles that define their product identity. By creating a product board, a wide variation of Husqvarna products was studied to gain knowledge of what defines their unique design. During the design ideation process, specific products were studied to gain inspiration for the concept that was developed. This has made the basis of the project owners' idea of what defines the product identity.

Husqvarna's use of colour is a big reason why the brand is so easy to recognize. Most of their products use the same colours, which help the customer immediately associate the product with Husqvarna, even from a distance. The combination of matte dark grey and shiny orange creates a bold look that stands out anywhere. This colour choice not only grabs attention but also supports the idea that these are tough, high-quality products.

Beyond the colours, Husqvarna's design integrates sharp, clean lines and purposeful contours that give the products a modern and professional look. The design communicates both innovation, durability and functionality, with a focus on ease of use and comfort. Keeping things simple is important, especially since many of the tools are made for professionals who need them to be easy to understand and efficient to use. Every part of the design, from how you hold the tool to how you use its features, is made to work well without giving up style.

6.1.2 Husqvarna robotic vacuum cleaner, Research question 2

The product is designed for professional users, which means it must demonstrate both functionality and durability. Even when performing long or demanding tasks, it should feel reliable and be easy to use. It is also developed to address a real problem, specifically the physical strain caused by repetitive manual cleaning in unergonomic positions. Since Husqvarna's original brand colours are an important part of the company's identity, the colour scheme stays true to it. These colours communicate confidence, quality, and reliability while helping users quickly identify the product.

Sharp edges and flowing curves are combined in the design to produce a modern yet robust appearance. This balance gives the product a strong, clean look that complements Husqvarna's visual style. Additional contours are integrated into larger surface areas to add depth and visual interest, enhancing the overall aesthetic while reinforcing the brand's signature design approach.

A lot of the design is inspired by Husqvarna's large robotic lawn mower, Ceora, which does a good job of showing the brand's design language. Just like Ceora, the vacuum robot is split into two main parts, which makes the layout easy to understand and adds a smart and balanced feeling. This design also gives the product a more advanced and high-tech appearance, while keeping it practical.

To control the cleaning process, the product includes a display with access to the most important functions. The layout and look of the display are based on other Husqvarna machines, so it feels familiar to users who have used their tools before. Buttons and menus are made simple and clear to make the product easy to use, even for those who may not be used to robotic tools. The goal was to make a vacuum that feels professional, easy to work with, and true to Husqvarna's design and values. The results were inspired as well as compared to the analysis of Husqvarna's form language as shown in *Figure 34*.

The final design was also evaluated by the supervisor and representatives from Semcon to determine if Husqvarna's design language had been successfully achieved and the feedback confirmed that this goal was met.



Figure 34: Comparison of Husqvarna's form language and the final concept

6.2 Used methods

All methods used in the project were important for reaching the final concept. Each step gave useful insights and helped shape a solution suited to the environment it was designed for.

The interview early in the process was very helpful in finding out about real problems in the target setting. It gave direct information that couldn't have been found just by reading, and helped make sure the concept was based on real user needs.

Since both team members had done market research together, the brainstorming session became a bit limited. They had similar ideas, which made it harder to come up with a wide range of solutions. Still, the session showed how important it is to include all kinds of ideas, even ones that seem unusual.

Sketches were made to explain and build on early ideas. This was an important step before using digital tools, as it helped show different directions and improve the concepts.

When working with CAD to model the surfaces of the final concept, the team had to learn how to use the software properly. Since they were new to this kind of modelling, it took time and practice to get the shapes right and meet the design goals.

Creating realistic images of the concept also brought challenges. It took a lot of effort to make the renderings both lifelike and clear enough to show how the product works. These images were key in showing the idea to others.

The prototype took time to finish, especially to get the surfaces to feel right. But it was an important step to test the design in real life and make any final changes.

Throughout the project, the team learned a lot and gained useful experience. These lessons will be valuable in future work. Being able to solve design problems, explain ideas clearly, and use new tools are important skills that can be used in many types of jobs and projects in the design field.

6.3 Further development

The product's modular design offers a special chance to develop a variety of features that will make it extremely versatile for the future development of a robotic vacuum cleaner meant for warehouse environments. A modular approach enables customization based on the needs of the user by allowing the installation of various attachments or components designed for specific tasks.

One potential direction for development is adding a front attachment for mopping floors. This would expand the vacuum's functionality, making it suitable for various floor types in a warehouse setting, where both cleaning and mopping might be required. Additionally, creating a version of the product designed for outdoor use would be another valuable addition. This could involve using more rugged materials, weather-resistant components, and features suited for cleaning outdoor surfaces like pavements, factory yards, or construction areas.

Within this project, further exploration of detailed features could have brought the product closer to the manufacturing stage. For example, incorporating hinges for an access door or compartment would allow for easier maintenance and cleaning of internal parts. More consideration could have been given to how dirt and debris are managed within the robot, ensuring efficient disposal and preventing clogging in the system.

Designing the charging station and debris disposal station could be a focus point of future development. In addition, the charging station's design could be improved to blend in perfectly with a warehouse setting by adding features that are appropriate for the setting. To ensure that it supports the robot's functionality in industrial settings, it could be made resistant to dust and dirt, more space-efficient, and more durable.

To ensure that the final design truly aligns with Husqvarna's core design principles, durable, innovative and functional, it would be valuable to apply Kansei Engineering as a validation tool. Kansei Engineering (KE) provides a structured method for translating user perceptions and impressions into measurable design elements. By systematically collecting user's emotional responses and linking them to design features, KE can help verify whether the design expresses these principles.

6.4 Conclusion

As a potential Husqvarna product, the project's results is a concept for an autonomous warehouse cleaning robot. The objective was to communicate the company's design language, which is centred on innovation, sustainability, and functionality, while investigating an original approach in a new market. The product's name, Pura, comes from a Latin word that means "pure" or "clean" and it symbolizes simplicity, quality, and uniqueness.

The robot's modular design allows it to clean both wide open spaces and small, difficult-to-reach places with interchangeable front modules. It can switch between modules with a smart docking system and runs independently, helping to ensure continuous operation even while undergoing maintenance. The robot reduces the need for human oversight, which may improve workplace safety and lower repetitive strain injuries among warehouse staff. Like Husqvarna's robotic lawn mowers, charging is done via a rear docking station, and a vacuum system at the same station fully automates trash disposal.

A user control panel on the robot allows users to personalize cleaning schedules, keep an eye on system health, and track progress in real time. LiDAR, ultrasonic sensors, and cameras works together to provide navigation, which allows the robot to map its surroundings, avoid obstructions, and return to its station when necessary. Additionally, it has a position light to notify surrounding workers of its presence in warehouse aisles for safety.

The design choices were guided by challenges identified through real interviews, aiming to create a safer and more comfortable work environment for warehouse cleaners, while also offering economic benefits for companies engaged in warehouse operations. The product is intended to reduce the risk of long-term injuries caused by physical strain and to support better ergonomic practices in the workplace.

Other colour schemes outside of the standard Husqvarna palette have been explored to help differentiate the product and match the needs of a new market. In the end a physical prototype that shows the design's form and proportions was made from PLA plastic for exhibitions and demonstrations.

7. References

[1] “Semcon,” Semcon, 2025. [Online]. Available: https://www.semcon.com/sv/om-oss/var-historia.

[2] “Husqvarna,” Husqvarna, 2025. [Online]. Available: https://www.husqvarna.com/se/utforska-och-upptack/husqvarnas-historia-over-325-ars-innovation/.

[3] K. T. Ulrish, S. D. Eppinger and M. C. Yang, “Product Development,” in *Product development and design*, 2020, pp. 11-35.

[4] K. Österlin, “Vad är design?,” in *Design i fokus*, p. 22.

[5] “Industrial Design Society of America,” 2025. [Online]. Available: https://www.idsa.org/about-idsa/advocacy/what-industrial-design/.

[6] K. Österlin, “Industridesign,” in *Design i fokus*, 2023, p. 37.

[7] medium, “Designing Design Methods,” 27 Feb 2016. [Online]. Available: https://medium.com/@interkatie/designing-design-methods-c3a1a9e188d3.

[8] Design Council, “The Doubble Dimond,” 2025. [Online]. Available: https://www.designcouncil.org.uk/our-resources/the-double-diamond/.

[9] T. Andersson, “Aesthetic Flexability - In Industriel Design Pratice,” 2021.

[10] T. Harris, “How Vacuum Cleaners Work,” howstuffworks, 16 April 2024. [Online].

[11] J. Layton, “How Robotic Vacuums Work,” howstuffworks, [Online]. Available: https://electronics.howstuffworks.com/gadgets/home/robotic-vacuum.htm.

[12] Aeno, “Robot Vacuum Cleaner History: Who Made the First Robot Vacuum,” 2025. [Online]. Available: https://aeno.com/blog/robot-vacuum-cleaner-history-who-made-the-first-robot-vacuum/.

[13] Vacuumwars, “A Brief History of the Robot Vacuum Cleaner,” 28 August 2024. [Online]. Available: https://vacuumwars.com/history-of-the-robot-vacuum-cleaner/.

[14] VacuumWars, “The Saros Z70: Roborocks Bold Leap into Robotic Arm Technology,” 9 January 2025. [Online]. Available: https://vacuumwars.com/the-saros-z70-roborocks-bold-leap-into-robotic-arm-technology/.

[15] motioncm, 2024. [Online]. Available: https://motioncm.co.uk/the-evolution-of-industrial-cleaning-machines/.

[16] S. Malik, "Lidar SLAM: The Ultimate Guide to Simultaneous Localization and Mapping," wevolver, 04 03 2023. [Online]. Available: https://www.wevolver.com/article/lidar-slam.

[17] F. Maroto, "Sensors in Autonomous Mobile Robots for Localization and Navigation," techtarget, 2021. [Online]. Available: techtarget.com.

[18] M. Oitzman, "Navigating Positioning Technology for Autonomous Robot Design," the robot report, 2021. [Online]. Available: therobotreport.com.

[19] Sweepscrub, "Best way to clean a warehouse floor," 2025. [Online]. Available: https://sweepscrub.com.au/news/2021/12/31/best-way-to-clean-a-warehouse-floor.

[20] kearcher, "Warehouse Cleaning," 2025. [Online]. Available: https://www.kaercher.com/int/professional/know-how/warehouse-cleaning.html.

[21] makita, "Robotdammsugare LXT," 2025. [Online]. Available: https://www.makita.se/product/drc200.html.

[22] pudurobotics, 2025. [Online]. Available: https://www.pudurobotics.com/en/products/mt1.

[23] hako, "Scrubmaster B400 RH," [Online]. Available: https://www.hako.se/produkter/scrubmaster-b400-rh/.

[24] K. T. Ulrish, S. D. Eppinger and M. C. Yang, Product Development and Design, 2020.

[25] K. T. Ulrish, S. D. Eppinger and M. C. Yang, "Identifying Customer Needs," in *Product Development and Design*, 2020, pp. 77-94.

[26] K. T. Ulrish, S. D. Eppinger and M. C. Yang, "Product Specifications," in *Product Development and Design*, 2020, pp. 95-120.

[27] K. T. Ulrish, S. D. Eppinger and M. C. Yang, "Concept generation," in *Product Design and Development*, 2020, p. 132.

[28] K. T. Ulrish, S. D. Eppinger and M. C. Yang, "Concept Selection," in *Product Design and Development*, 2020, pp. 149-167.

[29] K. T. Ulrish, S. D. Eppinger and M. C. Yang, "Concept testing," in *Product Design and Development*, 2020, pp. 171-186.